

Technology Stack

Date	08 October 2026
Team ID	SWTID-2026-2386
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, vanilla JavaScript, Jinja2 template	No build step and fast to load; AI text is rendered with <code>textContent</code> to prevent injection; responsive on phones
2	Backend / Server-Side	Python 3, FastAPI, Uvicorn, Pydantic	Simple, automatic request validation and API docs at <code>/docs</code> ; easy for the whole team to read
3	Database / Data Storage	None	EduGenie is stateless; nothing is stored. A database is listed as a future enhancement (progress tracking)
4	Cloud / Hosting / Deployment	Local Uvicorn server, Docker (<code>Dockerfile</code> provided)	Simple to run for the demo; the container image can be deployed to Render, Railway or any cloud VM
5	Version Control & CI/CD	Git and GitHub, pytest	Collaboration for five members; automated test suite ready to plug into GitHub Actions
6	Third-Party APIs / Other Tools	Google Gemini API (<code>google-generativeai</code> SDK), python-dotenv	Gemini generates answers, explanations, summaries, quizzes and learning paths; dotenv keeps the key out of the code